HDOC DAY-12 ACTIVITY REPORT

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Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform an operation based on the user's choice.

The operations are: (1) find the sum and product of the first and last digits, (2) count digits lesser than, equal to, and greater than a given digit k, and (3) exit.

Algorithm

1. Start.
2. Input a positive integer n.
3. Display a menu with Choice 1, Choice 2, and Exit.
4. Read the user's choice.
5. For Choice 1, store the last digit using n % 10. Use a temporary variable and repeatedly divide it by 10 until the first digit remains. Calculate and display the sum and product of the first and last digits.
6. For Choice 2, input digit k. Initialize lesser, equal, and greater counters to 0. Check every digit of n and increment the appropriate counter depending on whether the digit is less than, equal to, or greater than k. Display the three counts.
7. For Choice 3, terminate the program.
8. For an invalid choice, display an error message.
9. Repeat the menu until Exit is selected.
10. Stop.

Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	n = 12345 Choice = 1	First digit = 1 Last digit = 5 Sum = 6 Product = 5	First digit = 1 Last digit = 5 Sum = 6 Product = 5	Pass
2	n = 12345 Choice = 2 k = 3	Count_lesser = 2 Count_equal = 1 Count_greater = 2	Count_lesser = 2 Count_equal = 1 Count_greater = 2	Pass
3	Choice = 3	Program terminated.	Program terminated.	Pass

Program Code - Screenshots

```
UW PICO 5.09                                  File: day12.cp

#include <stdio.h>

int main()
{
    int n, choice, k;
    int first, last, sum, product;
    int digit, lesser, equal, greater;
    int temp;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    do
    {
        printf("\n===== MENU =====\n");
        printf("1. Sum and Product of First and Last Digits\n");
        printf("2. Count Digits Lesser, Equal and Greater than k\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                temp = n;

                last = temp % 10;

                while (temp >= 10)
                {
                    temp = temp / 10;
                }

                first = temp;

                sum = first + last;
                product = first * last;

                printf("First digit = %d\n", first);
                printf("Last digit = %d\n", last);
                printf("Sum = %d\n", sum);
                printf("Product = %d\n", product);

                break;

            case 2:
                printf("Enter digit k (0-9): ");
                scanf("%d", &k);

                lesser = 0;
                equal = 0;
                greater = 0;
                temp = n;

                while (temp > 0)

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```

UW PICO 5.09 File: day12.cp

```
                while (temp >= 10)
                {
                    temp = temp / 10;
                }

                first = temp;

                sum = first + last;
                product = first * last;

                printf("First digit = %d\n", first);
                printf("Last digit = %d\n", last);
                printf("Sum = %d\n", sum);
                printf("Product = %d\n", product);

                break;

            case 2:
                printf("Enter digit k (0-9): ");
                scanf("%d", &k);

                lesser = 0;
                equal = 0;
                greater = 0;
                temp = n;

                while (temp > 0)
                {
                    digit = temp % 10;

                    if (digit < k)
                        lesser++;
                    else if (digit == k)
                        equal++;
                    else
                        greater++;

                    temp = temp / 10;
                }

                printf("Count_lesser = %d\n", lesser);
                printf("Count_equal = %d\n", equal);
                printf("Count_greater = %d\n", greater);

                break;

            case 3:
                printf("Program terminated.\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 3);
```

^G Get Help ^O WriteOut ^R Read File ^Y Prev Pg ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where is ^V Next Pg ^U UnCut Text ^T To Spell

UW PICO 5.09 File: day12.cp

```
                digit = temp % 10;

                if (digit < k)
                    lesser++;
                else if (digit == k)
                    equal++;
                else
                    greater++;

                temp = temp / 10;
            }

            printf("Count_lesser = %d\n", lesser);
            printf("Count_equal = %d\n", equal);
            printf("Count_greater = %d\n", greater);

            break;

        case 3:
            printf("Program terminated.\n");
            break;

        default:
            printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 3);

    return 0;
}
```

^G Get Help ^O WriteOut ^R Read File ^Y Prev Pg ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where is ^V Next Pg ^U UnCut Text ^T To Spell

Compilation, Execution and Testing

```
[bhavya@Bhavyas-MacBook-Air ~ % nano day12.cp
[bhavya@Bhavyas-MacBook-Air ~ % clang day12.cp -o day12
[bhavya@Bhavyas-MacBook-Air ~ % ./day12
Enter a positive integer: 12345

===== MENU =====
1. Sum and Product of First and Last Digits
2. Count Digits Lesser, Equal and Greater than k
3. Exit
Enter your choice: 1
First digit = 1
Last digit = 5
Sum = 6
Product = 5

===== MENU =====
1. Sum and Product of First and Last Digits
2. Count Digits Lesser, Equal and Greater than k
3. Exit
Enter your choice: 2
Enter digit k (0-9): 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2

===== MENU =====
1. Sum and Product of First and Last Digits
2. Count Digits Lesser, Equal and Greater than k
3. Exit
Enter your choice: 1
First digit = 1
Last digit = 5
Sum = 6
Product = 5

===== MENU =====
1. Sum and Product of First and Last Digits
2. Count Digits Lesser, Equal and Greater than k
3. Exit
Enter your choice: 
```

Result

The program was compiled and executed successfully. For the tested number 12345, Choice 1 produced Sum = 6 and Product = 5. For k = 3, Choice 2 produced Count_lesser = 2, Count_equal = 1, and Count_greater = 2. The actual outputs match the expected outputs.